# Time Tokens

*Making Time Visible, Together*

Study Exercise Gaming

## *Project Overview*

Time Tokens is a tangible time-awareness system connecting three activity tokens—Study, Exercise and Gaming—with a shared Time Tower. Users record intentional one-hour sessions through physical interaction. Completed time units become coloured activity traces on the Tower, creating opportunities for individual and shared reflection.

01 — From Problem to Opportunity

Starting with Combatting Computer Game Addiction, our research shifted towards time allocation and task prioritisation.

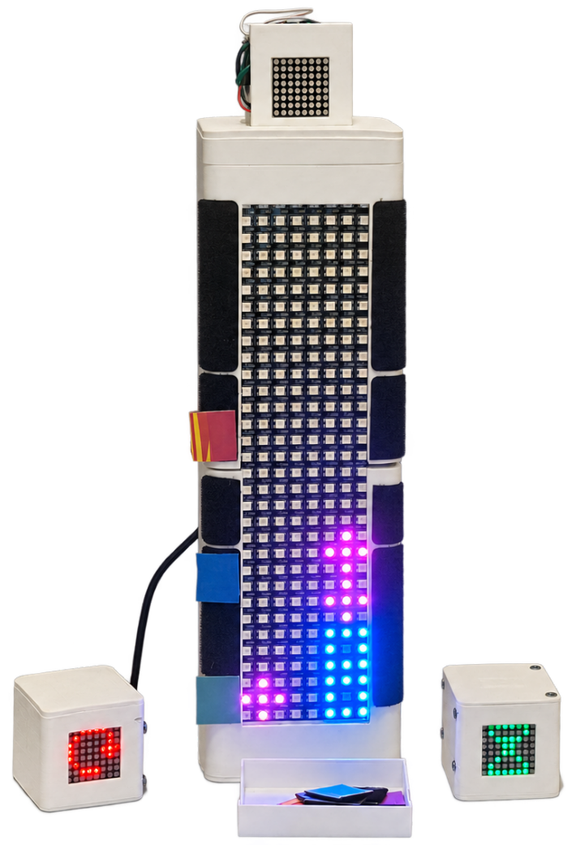
Rather than restricting gaming, we explored how physical time recording could help people notice their routines and reflect on everyday choices.

Design Question

*How might we make personal time use visible and encourage reflection without judging or restricting people's activities?*

LATEST BUILD

Plan markers
+ activity traces



HOOK-AND-LOOP PLANNING STRIPS

### *Social — Shared Activity Traces*

Separate columns make individual contributions visible, offering a common reference for noticing and discussing daily routines.



TEST PHOTO / STYLISED

### *Mobile — Activity-Based Contexts*

Personal Tokens are designed to support recording within different activity settings, connecting individual routines with a shared display.



TEST PHOTO / STYLISED

02 — The Designed Experience

## Flip → Track → Transfer → Reflect & Restart



FLIP PROMPT

### *01 Choose & Flip*

Select Study, Exercise or Gaming and flip the Token to begin.



TIMING

### *02 Track*

LED patterns show progress through a one-hour session, with feedback at completion.



TRANSFER SUCCESS

### *03 Transfer*

Completed time units are automatically sent to the shared Tower.



EARLIER TOWER DISPLAY

### *04 Reflect & Restart*

View accumulated records and actively choose whether to begin another session or change activities.

## *What We Learned*

### *01 Visibility Needs Clarity*

EVIDENCE

Users interpreted accumulated time, but some symbols and user identities were unclear.

INSIGHT

Seeing a record is not the same as understanding it.

P01-021 · P02-036-040

### *02 Sharing Has Social Trade-offs*

EVIDENCE

Different preferences for privacy and time detail emerged, alongside comparison in staged conversations.

INSIGHT

More information may help interpretation but increase social exposure.

P01-042 · P02-026-027, P02-041-043

### *03 Interaction Depends on Context*

EVIDENCE

Exercise raised portability concerns; uncertain feedback led to repeated flipping when the Tower was not visible.

INSIGHT

Tokens need understandable local feedback and appropriate physical mobility.

P01-055-062 · P02-066

04 — Design Insights & Decisions

### *Reflection vs. Interruption*

Earlier feedback raised concerns about hourly restarting. We deliberately retained the one-hour active checkpoint to invite users to reconsider their next activity. Its effectiveness for reflection remains unverified.

### *Clarity, Privacy & Mobility*

Evaluation identified opportunities to improve activity labels, record ownership, Token feedback and portability, while balancing information clarity with privacy.

05 — Reflection & Next Steps

Time Tokens visualises recorded activity time—not productivity. Further real paired testing and longer contextual use are needed to examine social interaction, privacy, hourly interruptions and incomplete-hour recording.

Explore our research, prototype and evaluation:

Yucheng Chen / Yiyang Qin / Bo Wang / Zhaoguo Huang / Yuming Zhong
Cutouts and test-photo illustrations: AI-assisted. Original photos retained in project archive.